```
A = [1 2; 3 4];
B = [5 6; 7 8];

C = A * B;

disp(C)
```

```
    19    22
    43    50
```

```


n = 5;
fact = 1;

for i = 1:n
    fact = fact * i;
end

disp(fact)
```

```
   120
```

```


n = 10;

a = 0;
b = 1;

disp(a)
```

```
     0
```

```
disp(b)
```

```
     1
```

```


for i = 3:n
    c = a + b;
    disp(c)

    a = b;
    b = c;
end
```

```
     1

     2

     3

     5

     8
```

```
    13

    21

    34
```

```
n = 17;
flag = 1;

for i = 2:n-1
    if rem(n,i)==0
        flag = 0;
        break;
    end
end

if flag==1
    disp('Prime Number')
else
    disp('Not Prime Number')
end
```

```
Prime Number
```

```
num = 14;

if rem(num,2)==0
    disp('Even Number')
else
    disp('Odd Number')
end
```

```
Even Number
```

```
num = 1234;
reverse = 0;

while num > 0
    digit = rem(num,10);
    reverse = reverse*10 + digit;
    num = floor(num/10);
end

disp(reverse)
```

```
        4321
```

```
A = [10 20 30 40 50];
```

```
sum = 0;

for i = 1:length(A)
    sum = sum + A(i);
end

disp(sum)
```

```
   150
```

```
A = [10 20 30 40 50];
key = 30;

found = 0;

for i = 1:length(A)

    if A(i)==key
        found = 1;
        position = i;
        break;
    end

end

if found==1
    disp(['Element Found at Position ',num2str(position)])
else
    disp('Element Not Found')
end
```

```
Element Found at Position 3
```

```
A = [5 2 8 1 4];

n = length(A);

for i = 1:n-1

    for j = 1:n-i

        if A(j)>A(j+1)

            temp = A(j);
            A(j)=A(j+1);
            A(j+1)=temp;

        end
```

```
    end

end

disp(A)
```

```
    1     2     4     5     8
```

```
str = 'madam';

if strcmp(str, flip(str))
    disp('Palindrome')
else
    disp('Not Palindrome')
end
```

```
Palindrome
```